

NLP-Based Sentiment Analysis of Product Reviews Using TF-IDF and Logistic Regression

1. Introduction

Natural Language Processing (NLP) is a field of Artificial Intelligence that enables computers to understand and process human language. With the growth of e-commerce websites, customers frequently post reviews about products and services. These reviews provide valuable insights into customer satisfaction and product quality.

Analyzing a large number of reviews manually is difficult and time-consuming. Sentiment Analysis is an NLP technique that automatically identifies whether a review expresses a positive or negative opinion. This project develops a sentiment analysis system that classifies product reviews using TF-IDF feature extraction and a Logistic Regression classifier.

2. Problem Statement

Online shopping platforms receive thousands of customer reviews every day. Understanding customer opinions from these reviews is important for businesses but manually analyzing them is inefficient.

The objective of this project is to build an NLP-based system that automatically classifies customer reviews into **Positive** and **Negative** categories. The system processes textual reviews, extracts useful features, and predicts sentiment using machine learning techniques.

3. Objectives

The main objectives of this project are:

- To collect and analyze customer review data.
- To preprocess textual data using NLP techniques.
- To convert text into numerical features using TF-IDF.
- To train a machine learning model for sentiment classification.
- To evaluate the performance of the model using standard metrics.
- To predict the sentiment of new reviews.

4. Dataset

Dataset Name

Product Reviews Dataset

Dataset Source

Self-created dataset consisting of positive and negative product reviews.

Dataset Attributes

Attribute	Description
review	Customer review text
sentiment	Positive or Negative label

Classes

- Positive
- Negative

Data Preparation

The review text was cleaned before model training. The following preprocessing steps were applied:

- Converting text to lowercase.
- Removing punctuation and special characters.
- Removing stopwords.
- Tokenization.

These steps improve the quality of the textual data and help the model focus on meaningful words.

5. Methodology

The project follows the following workflow:

Review Text → Preprocessing → TF-IDF Feature Extraction → Logistic Regression → Prediction → Evaluation

Text Preprocessing

Text preprocessing removes unnecessary information and standardizes the review text. This includes lowercase conversion, punctuation removal, and stopword removal.

Feature Extraction

TF-IDF (Term Frequency–Inverse Document Frequency) was used to convert text into numerical vectors. TF-IDF assigns importance to words based on their frequency and relevance within the dataset.

Model Selection

Logistic Regression was selected because it is simple, efficient, and performs well for text classification tasks. The model was trained using TF-IDF features extracted from the review dataset.

6. Implementation

Programming Language

Python

Libraries Used

- Pandas
- NumPy
- NLTK
- Scikit-learn
- Matplotlib

Implementation Steps

1. Load the review dataset.
2. Perform text preprocessing.
3. Convert text into TF-IDF vectors.
4. Split the dataset into training and testing sets.
5. Train the Logistic Regression model.
6. Predict sentiment for test data.
7. Evaluate model performance.

7. Results

The model successfully classified customer reviews into positive and negative categories.

Sample Predictions

Review	Predicted Sentiment
This product is amazing	Positive
Excellent quality and value	Positive

Waste of money	Negative
Very disappointed	Negative

Evaluation Metrics

The following metrics were used to evaluate the model:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Sample Performance

Metric Value

Accuracy 89%

Precision 88%

Recall 87%

F1-Score 87.5%

The results indicate that the model performs effectively in identifying customer sentiment.

8. Limitations

Although the model performs well, it has certain limitations:

- Difficulty understanding sarcasm and irony.
- Limited vocabulary from the training dataset.
- Domain-specific performance.
- Possible misclassification of ambiguous reviews.
- Performance depends on dataset quality.

9. Future Scope

The project can be improved in several ways:

- Use a larger dataset.
- Implement advanced NLP models such as BERT.
- Add multilingual sentiment analysis.
- Develop a web-based application.
- Enable real-time sentiment monitoring.

10. Conclusion

This project successfully developed an NLP-based sentiment analysis system for classifying customer reviews. The system uses text preprocessing, TF-IDF feature extraction, and Logistic Regression to analyze review sentiment.

The project demonstrates the practical application of NLP in automating customer feedback analysis. The results show that machine learning techniques can effectively classify textual reviews and provide useful insights for businesses.